

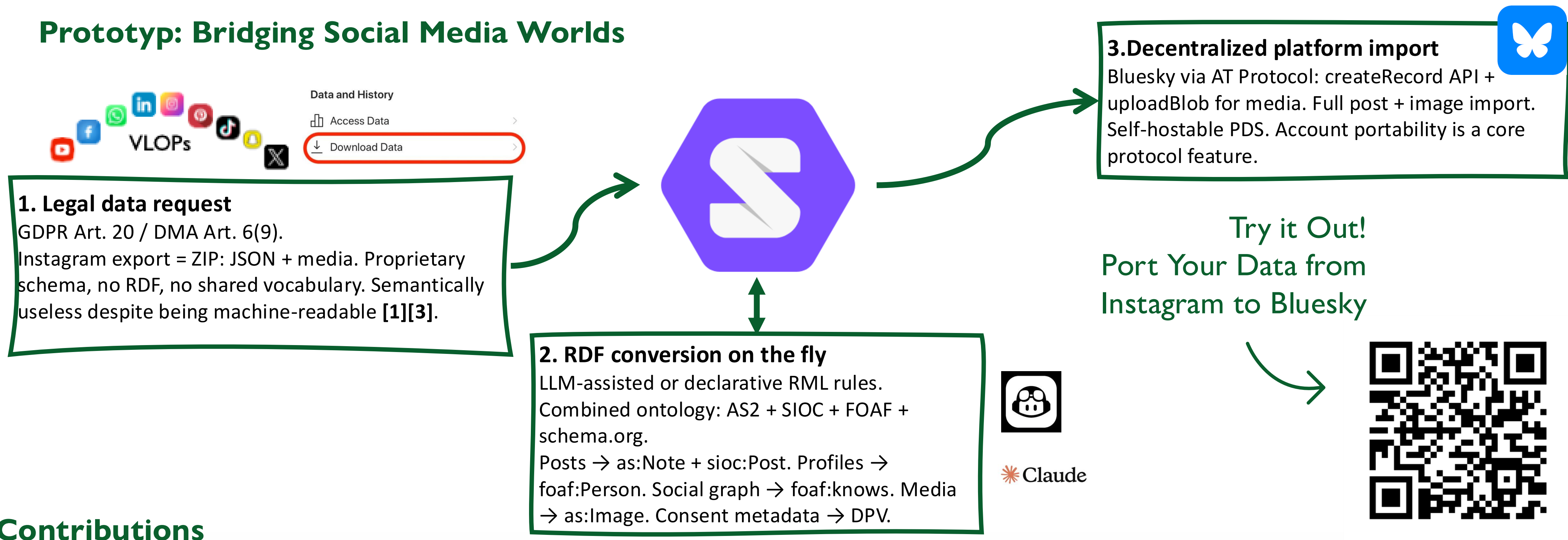
From Data Portability to Data Control: Making Regulation Technically Real

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Problem: European regulation (GDPR Art. 20, DMA Art. 6(9)) mandates social media data portability, but platform exports are proprietary data dumps with no semantic structure — technically compliant, practically hard to use for switching.

Approach: We build an end-to-end pipeline that uses LLM-assisted RDF conversion to turn platform exports into structured linked data hosted in user-controlled Solid Pods, enabling cross-platform import to decentralized services like Bluesky.

Prototyp: Bridging Social Media Worlds



Contributions

1. Working end-to-end pipeline:

Platform JSON data export. → RDF → Solid Pod → Decentralized platform import

2. LLM-assisted ontology mapping — Via an agent skill a LLM coding agent converts proprietary JSON to AS2/SIOC/FOAF/schema.org RDF [4] without pre-agreed schemas, reducing format conversion cost to near zero

3. Solid as the inter-protocol bridge — the Pod serves as a protocol-agnostic semantic data store between AT Protocol, ActivityPub, and centralized platforms

What do we do with this Info?

For regulation

The problem is not the export *format* but data *completeness* and *access*. Mandating a standardized export schema means a political negotiation. Instead:
Mandate completeness — all user data, all metadata, no stripping, no omissions
Mandate documentation — published, versioned schemas so tools can parse exports
Mandate timeliness — real-time access
Leave the semantic layer to user-side infrastructure, not platform obligations

For the Solid ecosystem

- Solid Pod = protocol-agnostic semantic data store for Social Media platforms
- RDF as the canonical representation
LLMs translate *from* to any target format
- Solid enables portability between ecosystems

For the Interoperability community

- Data conversion cost → converges as demonstrated to ~0
- Remaining switching costs are mainly social (friends aren't on new platform) and behavioral (habit, filter bubble)
- Network effects weaken when "comcom" makes leaving cheap [2]

References and Prior Work this builds on:

- [1] Wong, J. & Henderson, T. (2018). How Portable is Portable? Exercising the GDPR's Right to Data Portability. In: Proc. UbiComp '18, pp. 911–920. ACM. → Empirically exercised data portability rights with real platforms. Found wide variation in formats, incomplete data, and practical barriers to reuse. Established that exports are technically machine-readable but practically unusable
- [2] Doctorow, C. (2021). Competitive Compatibility: Let's Fix the Internet, Not the Tech Giants. Communications of the ACM, 64(10), pp. 26–29. → Defined "competitive compatibility" (comcom) — plugging into incumbents' systems without their permission. Historically required reverse engineering. We show GDPR/DMA + LLMs achieve comcom through legal rights + AI.
- [3] De Hert, P., Papakonstantinou, V., Malgieri, G., Beslay, L. & Sanchez, I. (2018). . Computer Law & Security Review, 34(2), pp. 193–203. → Distinguished the minimalist "adieu scenario" (data dump on departure) from the empowering "fusing scenario" (semantic interoperability enabling genuine switching). Our pipeline implements the fusing scenario.
- [4] Breslin, J.G., Bojars, U. & Decker, S. (2010). Porting Social Media Contributions with SIOC. In: Recent Trends and Developments in Social Software, LNCS 6045, pp. 116–128. → Built SIOC exporters for WordPress, Drupal, phpBB, Twitter + a WordPress SIOC Import plugin. Demonstrated the complete export→RDF→import cycle. Closest conceptual predecessor — but predates Solid, GDPR, DMA, and all modern platforms.



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